

2024



Scroll Processes 15 Million Transactions with a Secure, Scalable Layer 2 Blockchain Solution on AWS

Scroll built its zkEVM 2 solution using Amazon EC2 and Amazon RDS, accelerating the adoption of Ethereum blockchain and incentivizing developers to build more services.

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200+
monthly projects delivered

2–3 weeks
saved during solution testing

15 million
transactions



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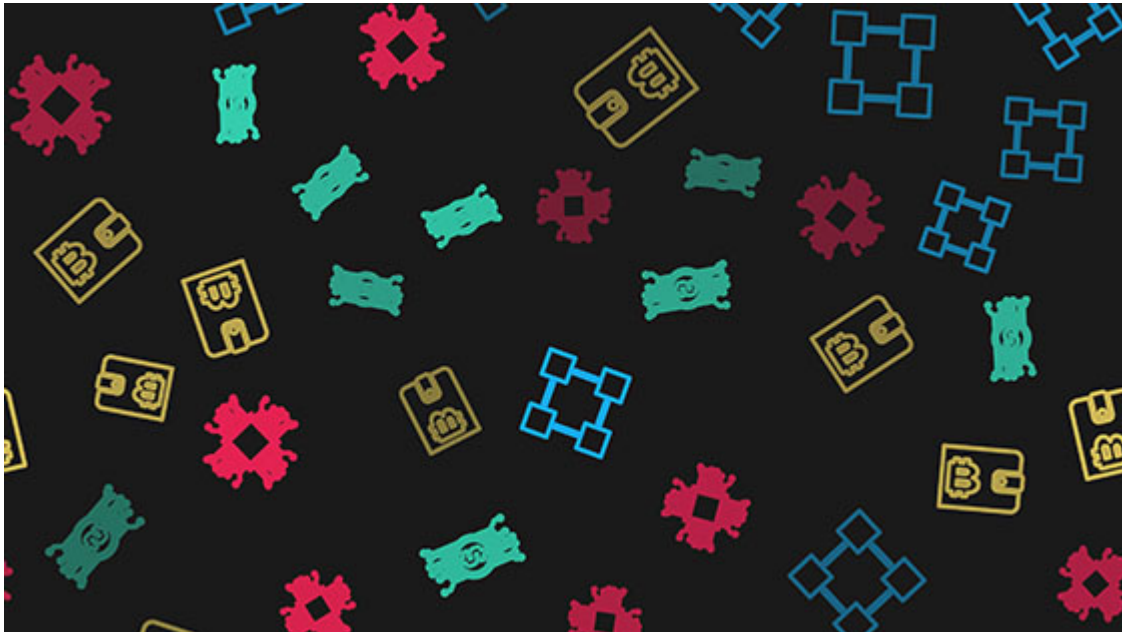
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Overview

[Scroll](#) is the developer of zkevm L2, a solution aimed at enhancing the performance of the Ethereum blockchain platform. The company hosts zkevm L2 on Amazon Web Services (AWS) for flexibility and security, using high-performance [Amazon Elastic Compute Cloud \(Amazon EC2\)](#), P4 instances with [Amazon Relational Database Service \(Amazon RDS\)](#) for database services and [AWS Security Hub](#) for security alerts.

By running zkevm L2 on AWS, Scroll is reducing latency and incentivizing developers to use Ethereum for their blockchain applications. The company has also reduced costs and saved weeks on zkevm L2 development, freeing up its team to focus more on innovation.



Opportunity | Accelerating the Adoption of Ethereum Blockchain

Hong Kong-based [Scroll](#) is driving the adoption of blockchain across industries, including healthcare, gaming, and finance. The emerging market for blockchain-based financial technology, known as decentralized finance (DeFi), is rapidly growing. A recent Statista report shows an [annual growth rate of 15.86 percent](#), with a projected total value of \$37 billion by 2028. To build applications for the DeFi market, developers use blockchain platforms like Ethereum to securely verify and execute transactions.

Scroll aimed to provide developers with improved transaction times across the Ethereum platform, addressing latency issues arising from Ethereum's focus on security and decentralization. To achieve this, the company integrates zero-knowledge technology with Ethereum virtual machines on the Layer 2 networking system—calling the resulting solution zkevm L2.

To run zkevm L2 on Ethereum's testnet and mainnet networks, Scroll needed a highly scalable and secure infrastructure. Haichen Shen, co-founder of Scroll, explains, "We sought a flexible infrastructure that offered a range of high-end GPUs to develop and launch our technology for the blockchain developer community to use."

Solution | Ensuring Scalable Blockchain Performance with Amazon EC2 P4 Instances

Scroll assessed multiple cloud solutions before choosing Amazon Web Services (AWS). “We appreciated the variety of CPU- and GPU-based options available on AWS, which gave us the right balance of price and performance,” says Shen.

Working closely with their local AWS team, Shen and his colleagues experimented with [Amazon Elastic Compute Cloud \(Amazon EC2\)](#) instances, scaling them across multiple AWS Regions.

The AWS team proposed deploying [AWS Organizations](#) to save time by managing AWS environments centrally. Scroll also opted to implement [AWS Security Hub](#), which automates security best-practice checks and aggregate security alerts, plus [AWS Identity and Access Management](#) to control access to AWS services.

In one instance, Scroll wanted to increase performance of the application’s [Amazon Relational Database Service \(Amazon RDS\)](#) for PostgreSQL. The AWS team shared information on [Amazon RDS Performance Insights](#), which Scroll used to fine-tune the database’s performance. “Our data engineers didn’t lose time thinking how to resolve performance challenges and could focus on other tasks,” states Shen.

To test zkevm L2’s performance ahead of its launch in September 2023, Scroll’s engineers ran 10 million test transactions across more than 2,000 Amazon EC2 instances, powered by 3rd Gen Intel Xeon Scalable processors. “The range of computing resources available on AWS saved a couple of weeks of testing, which helped us go to market that much quicker,” comments Shen.

By using AWS instead of an on-premises infrastructure for zkevm L2, Scroll has saved significant resources. Shen says, “We’re reducing our operational costs and passing on savings to our customers. Furthermore, we don’t need to maintain an infrastructure management resource, because we can launch servers easily ourselves.”

Outcome | Processing 15 Million Transactions and Supporting 200+ Projects

Since its launch on the mainnet, zkevm L2 has garnered adoption from hundreds of developers building Ethereum blockchain projects. AWS has played a pivotal role in processing over 15 million transactions and supporting their



various use cases, including DeFi, gaming, social applications, and non-fungible tokens (NFTs) or unique digital identifiers. In addition, developers are onboarding around 200 projects a month on the zkevm L2 solution.

Furthermore, Scroll is significantly contributing to accelerating the adoption of blockchain technology. By running zkevm L2 on AWS, the company is making transactions faster and more cost effective. This improvement incentivizes developers to create innovative solutions, driving the growth and potential of blockchain to transform a range of industries.

Scroll is also aiding the development of DeFi solutions that could benefit communities with limited access to traditional financial services. This includes crowdfunding solutions that provide community members with a secure means to collectively invest in local projects. “We’re very excited to be part of an industry that’s providing better access to potentially life-changing services through our zkevm L2 solution on AWS,” says Shen.

About Scroll

Scroll is a startup focused on improving the Ethereum blockchain platform. The company has developed its zkevm L2 solution on AWS to help reduce transaction times on the platform, improving the developer experience and driving Ethereum adoption.

AWS Services Used

Amazon EC2

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon RDS

Amazon RDS is an easy to manage relational database service optimized for total cost of ownership.

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AWS Organizations

AWS Organizations lets you create new AWS accounts at no additional charge.

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AWS Security Hub

AWS Security Hub is a cloud security posture management (CSPM) service that performs security best practice checks, aggregates alerts, and enables automated remediation.

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